Machine Learning Exercise Sheet 2

k-Nearest Neighbors and Decision Trees

Exercise sheets consist of two parts: In-class exercises and homework. The in-class exercises will be solved and discussed during the tutorial. The homework is for you to solve at home to further engage with the lecture content. You can upload your homework solutions in groups up to 4 to Moodle for a possible grade bonus. You do not have to upload any solutions of the in-class exercises.

In-class Exercises

There are no additional in-class exercises this week.

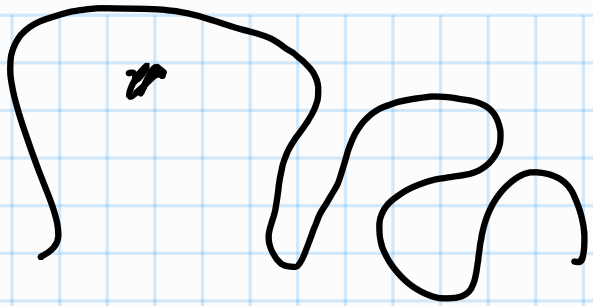
1 Homework

kNN Classification

Problem 1: You are given the following dataset, with points of two different classes:

Name	x_1	x_2	class
A	1.0	1.0	1
B	2.0	0.5	1
C	1.0	2.5	1
D	3.0	3.5	2
E	5.5	3.5	2
F	5.5	2.5	2

We perform 1-NN classification with leave-one-out cross validation on the data in the plot.

a) Compute the distance between each point and its nearest neighbor using L_1-norm as distance measure.

If you draw the points, you can identify the nearest neighbor without computing all distances: